Credit-risk monitor — solution overview

How a risk analyst uses it
  Portfolio: KPIs (accounts, exposure, default rate, predicted PD) then cuts by product, region,
  employment, income. Filters in the sidebar are the drill-down. Use this for morning book health.
  Account: paste an account_id. See PD, risk band, bureau/repay drivers, collections, score reasons.
  Anomalies: work the queue, read the why-flagged line, confirm or dismiss. Open items stay listed.
  Ask: type a question such as 'Which loan type has the highest default rate?' and get a cited figure.

AI feature 1 — Ask the Portfolio
  Trigger: analyst submits a question. Grounding: facts.json only. False positive: a well-formed
  question about a cut we did not pre-aggregate (we decline rather than invent). False negative:
  a phrasing we fail to parse locally when no API key is set. Human step: analyst checks the cited
  grouping against the chart on the Portfolio tab before acting.
AI feature 2 — Anomaly detection
  Trigger: peer-extreme bureau DPD/utilisation/write-off, missed EMIs, or IsolationForest tail.
  False positive: a large mortgage with a one-off utilisation spike that is still paying on time.
  False negative: a two-month-old account that has not yet missed. Human step: confirm/dismiss.

Model train 11,013 / test 3,672 accounts, stratified on target_default.
  Logistic AUC 0.668 KS 0.248 Gini 0.336
  LightGBM AUC 0.660 KS 0.240 Gini 0.321 ← production score
  Top drivers: loan_age_months, interest_rate_pct, loan_amount, years_employed, years_at_residence, annual_income.
  Monitor weekly PSI on income, loan amount, bureau DPD, write-off rate; monthly default vs PD band.

Top insights (1) Mortgage carries the highest default rate. (2) Unemployed customers are the weakest employment segme
  Book default rate 16.1%; weakest product Mortgage at 24.6%.

Assumptions Snapshot as-of 2026-10-31; missing bureau DPD = 0; missing rates imputed by product.
  target_default is the observed outcome, not a through-the-cycle PD. USD amounts as provided.
To production IT: scheduled CSV/warehouse extract, model registry, Streamlit/SSO, secrets vault,
  audit log of anomaly decisions. Data: monthly repayment tape, collections codes, official default
  definition, vintage and closed-account policy, region hierarchy, and a hold-out recent origination month.